

Use Case Specification

Site Doctor

Version 1.0

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1 Overview

This document specifies the primary use cases supported by Site Doctor, traceable back to the functional requirements defined in the SRS (v1.0). Each use case follows a standard structure: actor, preconditions, postconditions, main flow, alternative flows, and exception flows.

2 Actor Summary

Actor	Description
User	A website owner, freelancer, or developer running an audit against a target site. Provides check selections and fix approvals.
Site Doctor System	The orchestrated multi-agent pipeline; acts autonomously between human-in-the-loop checkpoints.
Lighthouse (external tool)	Invoked by the system to perform rule-based SEO/accessibility/performance auditing. Not a human actor, but modeled as a secondary actor since it is an independent process boundary.
LLM Provider (external service)	Invoked for triage, vision-based UX review, and fix generation.

3 UC-01: Analyze Website

Actor: User

Preconditions: User has a URL believed to be reachable. No prior audit run for this URL is required.

Postconditions (success): A report exists containing ranked Issues and/or UX Suggestions for the checks the user selected.

Postconditions (failure): No report is generated; the user receives a specific error explaining what failed.

Related Requirements: FR-01 through FR-20

Main Flow

1. User provides a target URL.
2. System validates the URL is well-formed and reachable.
3. System presents a check selection prompt (SEO / UX / Security), with Security defaulted to unselected.
4. User selects one or more checks.
5. System crawls and renders the page, always saving an HTML copy; captures scroll-position screenshots only if UX was selected.
6. System runs each selected audit branch (SEO / UX / Security) in parallel.

7. System waits for all selected branches to complete, then triages and ranks the combined findings.
8. System presents the final report, with Issues and UX Suggestions in clearly separated sections.

Alternative Flow A1: Only SEO Selected

At step 4, if the user selects only SEO, step 5 does not capture screenshots, and step 6 invokes only the `seo_audit` branch.

Alternative Flow A2: Security Explicitly Selected

At step 4, if the user explicitly selects Security, the system additionally confirms the user is authorized to test the target before proceeding to step 5 (see UC-04).

Exception Flow E1: Invalid URL

At step 2, if the URL is malformed, the system halts immediately with a validation error; no crawl or audit work is performed. (FR-02)

Exception Flow E2: Website Unreachable

At step 5, if the crawl times out or the target refuses connection, the system halts and returns a clear "unreachable" error rather than a raw stack trace. (SDD §8)

Exception Flow E3: External Tool Unavailable

At step 6, if Lighthouse or the LLM API is unavailable, the affected branch's findings are marked unavailable in the final report rather than failing the entire run; other selected branches still complete normally.

4 UC-02: Select Checks to Run

Actor: User

Preconditions: A valid URL has been provided (UC-01, step 1–2 complete).

Postconditions (success): `selected_checks` is populated with at least one of {seo, ux, security}.

Related Requirements: FR-03, FR-04

Main Flow

1. System presents SEO, UX, and Security as independently selectable options, with Security unchecked by default.
2. User selects any combination of options.
3. System records the selection and proceeds.

Exception Flow E1: No Check Selected

If the user submits with no checks selected, the system re-prompts rather than proceeding with an empty pipeline run.

5 UC-03: Review and Approve a Proposed Fix

Actor: User

Preconditions: A completed SEO audit run exists with at least one mechanically verifiable Issue for which the system has generated a proposed Fix.

Postconditions (success): Each proposed Fix has an explicit approved/rejected status; approved Fixes have been applied and re-verified.

Related Requirements: FR-21 through FR-26

Main Flow

1. System presents each proposed Fix with an explicit before/after diff and the Issue it addresses.
2. User approves or rejects each Fix individually (or via batch approval).
3. System applies only approved Fixes to the local HTML copy.
4. System re-runs the originating audit against the patched copy.
5. System marks each applied Fix as verified-cleared or not, and updates the report accordingly.

Alternative Flow A1: Fix Rejected

At step 2, if a Fix is rejected, steps 3–5 are skipped for that Fix; it remains listed in the report as an unresolved, human-reviewed Issue.

Alternative Flow A2: Fix Fails Verification

At step 5, if a Fix does not verify as cleared, the system routes back to fix generation for that Issue and repeats steps 1–5 for the new proposal, up to `max_retries_per_fix` attempts.

Exception Flow E1: Retry Limit Exhausted

If a Fix still fails verification after the configured maximum retries, the system marks the Issue as unresolved and retains the attempt history in the report, rather than looping indefinitely.

6 UC-04: Authorize and Run Security Check

Actor: User

Preconditions: User has explicitly selected the Security check during UC-02.

Postconditions (success): Passive security findings (headers, TLS configuration, exposed-version/CVE lookups) are included in the final report.

Related Requirements: FR-14, FR-15, FR-16

Main Flow

1. System requires the user to confirm they are authorized to test the target site before the security branch is permitted to run.
2. User confirms authorization.
3. System executes only passive checks against the target: HTTP security header inspection, TLS/certificate configuration review, and exposed software-version lookups against known-CVE sources.
4. System returns findings to the triage stage alongside any other selected branches.

Exception Flow E1: Authorization Not Confirmed

At step 2, if the user does not confirm authorization, the security branch is skipped entirely for this run; other selected branches proceed normally.

Design Constraint

Under no configuration does this use case perform active exploitation, authentication bypass attempts, or load/stress testing. This constraint is enforced at the routing level, not only presented as a UI warning. (SDD §9)

7 UC-05: View or Download Report

Actor: User

Preconditions: A completed audit run exists.

Postconditions (success): The user has viewed or obtained a copy of the report.

Related Requirements: FR-20

Main Flow

1. User requests the report for a completed run.
2. System renders the report, with Issues and UX Suggestions in separated sections, and any applied/verified Fixes noted.
3. User optionally downloads the report.

Exception Flow E1: Run Not Yet Complete

If the user requests a report for a run still in progress, the system returns the current status rather than a partial or empty report.